

Insertion Sort

unsorted array

7	4	6	5
0	1	2	3

Part of array as

- ① Sorted array
- ② unsorted array

In the beginning, the first element is considered as sorted array.

Step 1

Sorted	unsorted		
7	4	6	5
0	1	2	3

Includes one by one element from the unsorted array to a sorted array in a sorted fashion

Include 4; Key = 4;

Sorted		unsorted array	
7	/	6	5
0	1	2	3

$arr[i] > \text{Key}$ means do shift
 $7 > 4$ yes,

one index to the right

Sorted			
/	7	6	5

fill the space with Key value

Sorted	unsorted		
4	7	6	5

Include 6, key = 6;

Sorted	unsorted		
4	7	/	5

$7 > 6$, yes, do one shift right

4	/	7	5
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$4 > 6$, no. fill space with Key value

Sorted	unsorted		
4	6	7	5

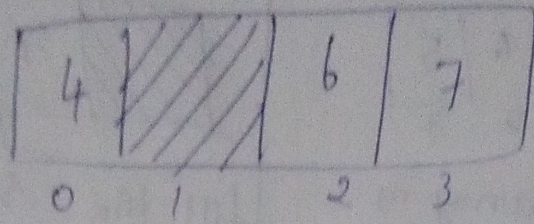
Include 5; Key = 5

4	6	7	/
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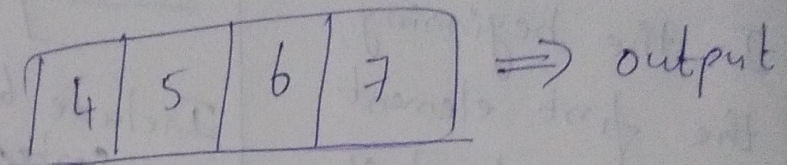
~~5~~ $7 > 5$; do one shift right

4	6	/	7
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$6 > 5$, yes, do one shift right



$4 > 5$; no, then replace key value
to space



Time Complexity

int ins_sort(int arr[]) {

{

n ← arr.length; ——— ①

int key; ——— ①

int space; ——— ①

for (int i = 1; i < n; i++)
{ ↳ $1 + n - 1 + 1 + n - 1 = 2n$ ②

key = arr[i]; ——— $2(n-1)$

space = i; (n-1)

for (int j = space - 1; j >= 0; j--)

{ ↳ $(n-1) + (x+1) + x$
 = $2x + n$

if (arr[i] > key) ——— (2x)

arr[j+1] = arr[j]; ——— (3x)

space = j; ——— (x)

Continue; ——— (x)

break; ——— X not performed

arr[space] = key; ——— (2(n-1))

return arr; ——— (1)

$$1 + 1 + 1 + 2n + 2n - 2 + n - 1 + 2x + n + 2x + 3x + x + x + 2n - 2 + 1$$

$$3 + 2n + 2n - 2 + n - 1 + 2x + n + 2x + 3x + x + x + 2n - 1$$

$$= 8n + 9x - 1$$

where $x = 1 + 2 + 3 + 4 + \dots + n - 1$

In A.P. $\frac{n(n+1)}{2}$

$$= \frac{n-1(n+1)}{2} = \boxed{\frac{n(n-1)}{2}}$$

$$= 8n + 9 \frac{n(n-1)}{2} - 1$$

$$= 8n + \frac{9n^2}{2} - \frac{9n}{2} - 1$$

∴ Time Complexity is $\boxed{O(n^2)}$

i	j	no. of times
1	0	1
2	1, 0	2
3	2, 1, 0	3
4	3, 2, 1, 0	4

$$= 1 + 2 + 3 + 4 + \dots + n - 1$$

$$= x \text{ times}$$